

TERM PAPER

On

SUSTAINABLE APPROACHES FOR LARGE LANGUAGE MODEL DEVELOPMENT

Submitted to

**AMITY SCHOOL OF ENGINEERING AND TECHNOLOGY
(ASET)**

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Declaration by the student

I, **Rashi Singh**, student of **B.Tech (CSE)** hereby declare that the project titled **“Sustainable Approaches for LLM Development”** which is submitted by me to Department of Computer Science and Engineering, **Amity School of Engineering and Technology (ASET), Noida**, Amity University Uttar Pradesh, in partial fulfilment of requirement for the award of the degree of Bachelor of Technology (Computer Science and Engineering), has not been previously formed the basis for the award of any degree, diploma or other similar title or recognition. The Author attests that permission has been obtained for the use of any copy righted material appearing in the report other than brief excerpts requiring only proper acknowledgement in scholarly writing and all such use is acknowledged.



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Noida

Date: 09/07/2024

CERTIFICATE

On the basis of report submitted by Name of student, student of **B.Tech Computer Science and Engineering**, I hereby certify that the report “**Sustainable Approaches for LLM Development**” which is submitted to Department of Computer Science and Engineering, **Amity School of Engineering and Technology (ASET), Noida**, Amity University Uttar Pradesh in partial fulfilment of requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering is an original contribution with existing knowledge and faithful record of work carried out by him/her under my guidance and supervision.

To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

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ABSTRACT

Large Language Models (LLMs) and Generative Artificial Intelligence (Gen AI) have grown rapidly, revolutionizing technology in fields such as science, engineering, research, and finance. Despite their huge benefits and crucial importance in a variety of industries, these improvements have created serious environmental concerns due to the carbon emissions produced by the models and the high-performance computing infrastructure required. The environmental impact of modern technologies cannot be overlooked, demanding immediate action to reduce their carbon footprint. This review paper addresses the urgency of the environmental damage caused by large language models, providing a comprehensive analysis of the issue and proposed solutions. These ideas aim to reduce carbon emissions while maintaining the quality and effectiveness of LLM services. The report also discusses alternate methodologies and tactics for future research and development that could reduce the environmental impact of LLMs. By investigating and adopting these methods, we can assure that the continuous development and deployment of large language models is both sustainable and environmentally responsible, balancing technical advancement with safeguarding the environment.

Table of Contents

S. No.	Topic	Page No,
1.	Introduction	6
2.	Methodology	9
3.	Observations	14
4.	Conclusion	17
5.	Discussion	18
6.	References	20

Tables and Figures

S. No.	Topic	Page No.
1.	Figure 1: Overview of System design of SPROUT	9
2.	Figure 2: Flowchart of the process used for testing and recording model performance	12
3.	Figure 3: Carbon savings achieved across all geographic regions	14
4.	Figure 4: Bar chart depicting the performance of models on various tasks from the GLUE benchmark	14
5.	Table 1: Language modelling tasks used for evaluation	11
6.	Table 2: Comparable performance of DistilBERT on downstream tasks	15
7.	Table 3: DistilBERT being significantly smaller and faster	15

INTRODUCTION

A. Definition and objective of the term paper

Large language models are artificial intelligence models that understand and produce human language. They are constructed using natural language processing and consist of a complex structure. Some prominent examples of these are ChatGPT by OpenAI, BERT, and T5 by Google. Although these chatbots provide great performance and help to the users, the increasing demand for GenAI is becoming a concern because of its environmental impact.

Large language models consume a huge amount of power, and this is directly proportional to the size of the datasets they are trained on. These models consist of parameters that are trained on a very large amount of data. The number of parameters directly affects the amount of data to handle and computations to make. And the process of training these models, which includes running complex algorithms on extensive amounts of data, requires an enormous amount of computational power and energy. This leads to various gas emissions and carbon footprints. The continuous requirement of electricity generation to facilitate the training processes could possibly contribute to 8% of global carbon emissions within a decade.

Training these models on energy-consuming CPUs and GPUs results in a huge electrical draw and also generates a large amount of heat. To compensate for the heat generated, some data centres use water-based cooling methods. But this method has its own disadvantages, as the water temperature rises, which impacts the local ecosystems and also pollutes the water used. Also, to avoid model drift, that is, decay in the predictive power of a model, some models need to be retrained frequently. These model maintenance practices lead to unnecessary energy and cost consumption. The data centres depend on fossil fuels for energy, which worsens the carbon emissions related to LLMs. Shifting towards some renewable energy sources can significantly help reduce the environmental impact of the large language models.

Green generative AI refers to the development and execution of artificial intelligence models with an emphasis on sustainable practices and energy efficiency in order to reduce their negative environmental effects. It aims for a balance between ecological responsibility and technological innovation, making sure that AI advances society without damaging the planet's health. Considering these issues, the purpose of this review paper is to draw attention to green generative AI by exploring possible sustainable approaches towards large language model development.

B. International and national status of the work in the area

Researchers and developers have realized the importance of addressing the environmental impact of large language models (LLMs), and they have made significant progress in eco-friendly AI globally.

Research Initiatives for Energy-Efficient Architecture: Several organizations are actively investigating novel model designs aimed at maintaining high performance while reducing energy usage. Techniques such as sparse neural networks and attention processes have shown promise in improving energy efficiency. **Green Data Centres and Renewable Energy Adoption:** Tech businesses are rapidly investing in green data centres that run on renewable energy sources such as solar, wind, and hydroelectric electricity. These initiatives are intended to lower the overall carbon footprint of AI operations.

Knowledge Distillation and Transfer Learning: Knowledge distillation reduces huge LLMs to smaller, more energy-efficient models while maintaining the knowledge. Transfer learning uses pre-trained models to minimize the training time for new models, saving energy. **Model Pruning and Compression Techniques:** Model pruning and compression approaches remove redundant parameters from an LLM, lowering its size and energy usage without drastically impacting performance. [5]

By implementing these initiatives, the AI community all over the world is aiming for more sustainable and environmentally responsible usage of large language models, balancing scientific profits with ecological concerns.

METHODOLOGY

The challenge therefore becomes, “How to reduce the environmental impacts of large language models without compromising service quality?” The answer involves various methods like generation directives, knowledge distillation, model compression, and parameter-efficient fine tuning. Many researchers have worked on these strategies throughout the years to develop green, generative AI systems. This is a thorough explanation of various possible approaches and how they lower carbon emissions without sacrificing the output quality for the client.

1. SPROUT: An environment friendly language model using Generation Directives

1.1 SPROUT is designed as a carbon-aware generative language model inference framework. It presents the concept of "generation directives," an approach to generating high-quality content while subtly adjusting the amount of auto-regressive inference iterations.

A generation directive is a set of instructions that, when combined with a prompt input, dictates the manner in which a generative language model produces tokens for the

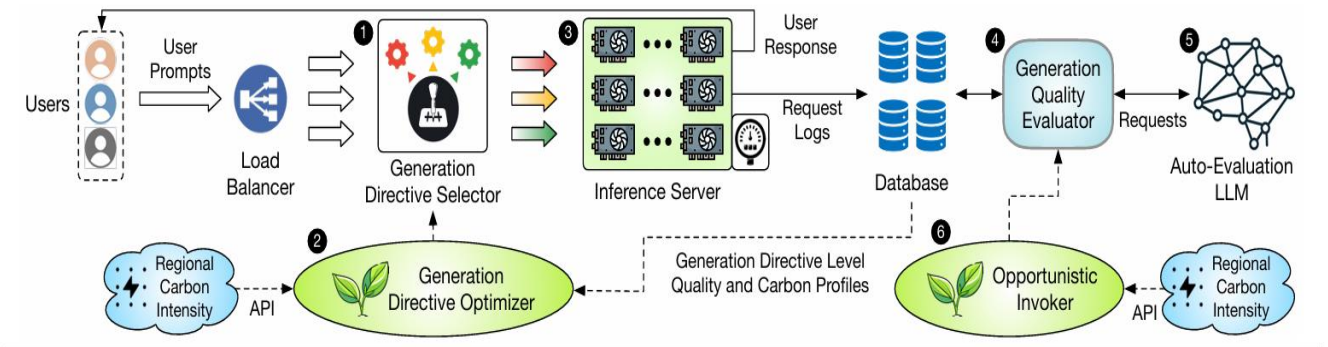


Figure 1: Overview of System design of SPROUT [1]

prompt. This guiding instruction is provided by a predefined text sequence that is specified for each generating directive level.

1.2 Design: To ensure high-quality content generation while reducing carbon footprints, SPROUT uses token generation directives. Once the load balancer has assigned the

users prompts to an inference server, they must be tokenized into numerical representations. During this stage, each prompt is given a directive level by a generation directive selector (1). The SPROUT's token generation directive optimizer (2) determines which policy to assign the directive levels to. After the autoregressive inference is completed on inference server (3), the resulting tokens are detokenized and returned to the user clients. After that, a generation quality evaluator (4) takes a sample of the database's prompts, creates responses for each at every level of the generation directive, and determines which level of the directive produces the best answer for each request. To assess generation feedback, an LLM-based automatic evaluator is used. In order to determine its preferences on the responses, this evaluator refers to an auto-evaluation LLM (5), logging these preferences back into the database. The entire procedure takes place offline, and SPROUT's opportunistic evaluation invoker (6) makes sure that assessments are only conducted when absolutely essential and during times of low carbon intensity. [1]

Generation Directive Optimizer: The optimizer of SPROUT is made to adapt dynamically to fluctuations in carbon intensity and the unpredictability of user-initiated tasks. In low-carbon intensity, it promotes directives that improve content quality, capitalizing on the lower carbon cost of creating new tokens. On the other hand, in situations with high carbon intensity, it chooses regulations that might marginally deteriorate quality while lowering emissions.

1.3 Experiment: Three levels of generation directives are used to evaluate SPROUT's efficiency: L0 is the default baseline, which has no directives; L1 is for "brief" generation; and L2 is for "very brief" generation. For offline quality assessments, GPT-4 serves as the auto-evaluation LLM, which can be accessed using the OpenAI API.

Table 1: Language modelling tasks used for evaluation [1]

Dataset	Description	Task
Alpaca [41]	Instructions generated by OpenAI's <code>text-davinci-003</code>	Instruction tuning
GSM8K [42]	Grade school math problems	Arithmetic and multi-step reasoning
MMLU [20]	Massive multitask language understanding	Multiple-choice questions
Natural Questions [43]	Real-user questions from Google	Question answering
ScienceQA [21]	School science subjects (e.g., Biology/Physics/Chemistry)	Multiple-choice science questions
TriviaQA [22]	Trivia questions collected by trivia enthusiasts	Reading comprehension

Table I lists the six different language modelling datasets that are used for evaluation. These datasets, which cover a wide range of domains and applications, are essential performance benchmarks for large language model leaders.

2 DistilBERT: a distilled version of BERT using Knowledge Distillation

2.1 A compression technique called knowledge distillation trains a compact model, the student, to replicate the behaviour of the larger model, the teacher, or a collection of models.

In student architecture, the student DistilBERT in this work has the same overall architecture as BERT. The number of layers is reduced by half, and the token-type embeddings—which function as miniature descriptions for each word, encapsulating its meaning and relationships to other words—as well as the pooler—which combines a series of embeddings into a sentence embedding—are eliminated.

2.2 Experiment: General Language Understanding: Using the General Language Comprehension Evaluation (GLUE) benchmark, a set of nine datasets for assessing natural language understanding systems, DistilBERT's language comprehension and generalization skills are evaluated. [6]

3 Assessment of various LLMs using Fine-Tuning and Role of CPUs and GPUs

3.1 This section assesses the carbon footprint of popular large language models, which have a high carbon footprint because of their enormous number of model parameters. In order to assist people in making practical attempts to reduce CO2 emissions when training their NLP models, this experimentation is being conducted to examine various strategies for doing so.

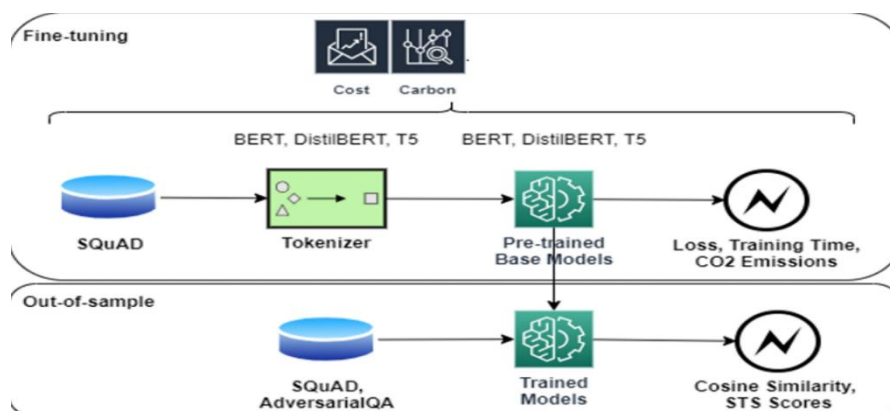


Figure 2: Flowchart of the process used for testing and recording model performance [3]

3.2 Experiment: Figure 1 illustrates the main phases of the experiment. In this project, various tokenizers and the SQuAD dataset are used to fine-tune huge language models, such as BERT, DistilBERT, and T5. Parameters including loss, training duration, and CO2 emissions are closely tracked during the fine-tuning stage. The effectiveness of the trained models is then evaluated using an out-of-sample test set that is obtained from the AdversarialQA and SQuAD datasets. Semantic Textual Similarity (STS) scores and cosine similarity are two evaluation metrics used in this context. [3]

3.3 Description of models used:

1) BERT, or Bidirectional Encoder Representations from Transformers, can be fine-tuned with a single output layer to create sophisticated designs for various tasks,

including question answering and language inference, without requiring substantial task-specific architecture modifications.

2) DistilBERT is a condensed version of the BERT model that is 60% faster and retains 97% of its language understanding skills. It was developed through knowledge distillation and is used in the pre-training stage of a BERT model's development.

3) T5, a transformer model in its fifth generation, can execute tasks given to it in a text-to-text manner.

4 Additional options to evaluate or adopt for sustainable Large Language models.

- i. Smaller models, with reduced training costs and lower environmental implications, perform almost as well. Stanford University researchers' Alpaca model and Meta AI's Large Language Model, or Llama, are both compact enough to operate on a desktop and can be built for hundreds rather than millions of dollars. [7]
- ii. One-shot or few-shot training is another method for lowering training expenses across the whole model lifecycle. One-shot training is a scenario in which a model is trained or fine-tuned using only one (or a few) instances for each class or task. Few-shot training includes training a model with a limited number of instances per class, which can range from a few to tens of examples. The goal is to get high performance using limited data.
- iii. Additionally, Corporations can improve the efficiency of their hardware by implementing different chip architectures or architecture tools based on that hardware. The SpiNNaker2 chip architecture, for example, simulates biological neural networks by enabling locally intensive processing over a sparsely active network. That is, where nothing is happening in the neural network, the chips use almost no power. Despite being created with bigger transistors, the chip architecture consumes far less power than most existing CPUs and GPUs while doing comparable computational tasks. [7]

OBSERVATIONS

I. From section (1), In a variety of geographical locations, SPROUT continuously achieves significant carbon savings while upholding criteria for generating quality. By using optimized generating instructions, SPROUT can cut carbon emissions by as much as 60%. All

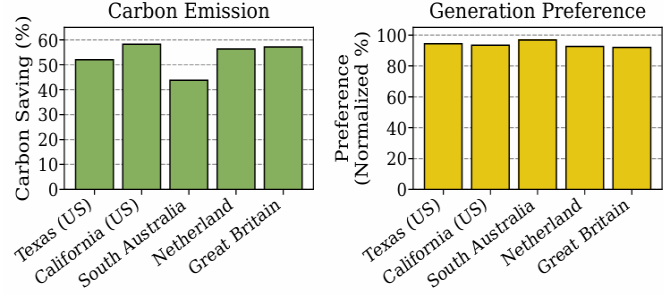


Figure 3: Carbon savings achieved across all geographic regions [1]

regions' generation preferences are still above 90%; in South Australia, they surpass 95% and save more than 40% of carbon. In the Californian Azure West US region, this corresponds to 3,266 tons of CO2 emissions each month. A monthly reduction of 1,903 tons of carbon could potentially be achieved by implementing SPROUT. [1]

SPROUT significantly decreases the carbon footprint related to LLM inference by utilizing the novel concept of generation directives. The assessment shows that SPROUT can, in a number of geographical regions, lower the carbon footprint of inference operations by more than 40%.

II.

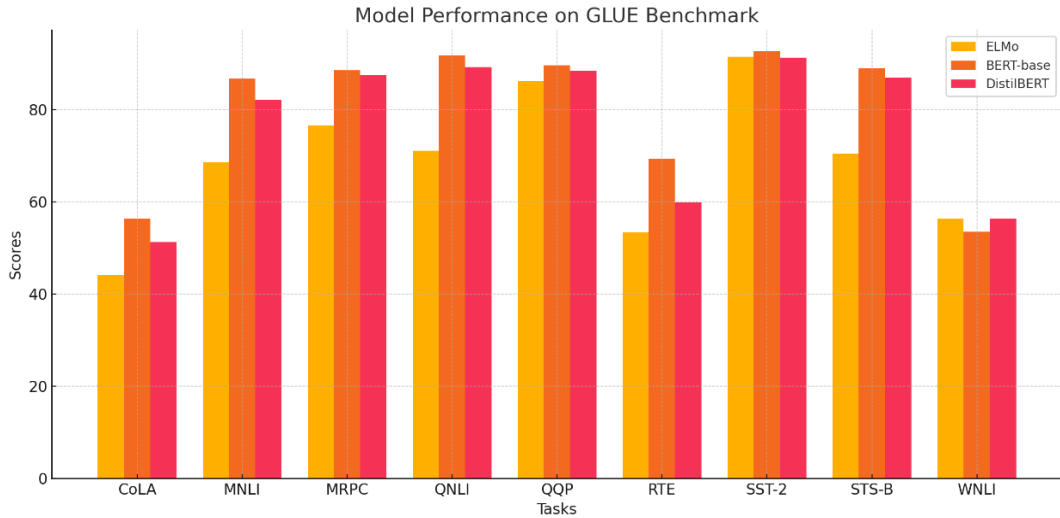


Figure4: Bar chart depicting the performance of ELMo, BERT-base, and DistilBERT on various tasks from the GLUE benchmark

From Section (2), Figure 4 displays the macro-score (average of individual scores) and the results for each of the nine tasks. DistilBERT consistently performs on par with or better than the Elmo baseline across all nine tasks. Remarkably well, DistilBERT retains 97% of the performance with 40% fewer parameters when compared to BERT. [6]

Benchmark for downstream tasks: Although being 40% smaller than BERT, DistilBERT only trails BERT by 0.6% points in test accuracy on the IMDb benchmark. DistilBERT is within 3.9 points of the complete BERT on SQUAD.

Table 2: Comparable performance of DistilBERT on downstream tasks

Model	IMDb (acc.)	SQuAD (EM/F1)
BERT-base	93.46	81.2/88.5
DistilBERT	92.82	77.7/85.8
DistilBERT (D)	-	79.1/86.9

Size and inference speed: Comparing the number of parameters of each model with the required inference time in order to better explore the speed-up/size trade-off of DistilBERT. DistilBERT is 60% faster and requires 40% fewer parameters than BERT.

Table 3: DistilBERT being significantly smaller and faster

Model	# param. (Millions)	Inf. time (seconds)
ELMo	180	895
BERT-base	110	668
DistilBERT	66	410

DistilBERT is a pre-trained general-purpose version of BERT that is 40% smaller and 60% faster than BERT while retaining 97% of its language understanding capabilities. We demonstrated how distillation can be used to successfully train a general-purpose language model and how an ablation study can be used to assess its many components. [6]

III. From Section (3), The experimental findings show that as compared to Distil BERT, the T5 and BERT models released significantly more CO₂. Moreover, employing a bert-base-cased tokenizer produced higher CO₂ emissions than using a distilbert-base-uncased tokenizer. The DistilBERT model with a distilbert-base-uncased tokenizer had 46.9% lower CO₂ emissions during model training with the T4 GPU than the BERT model with a bert-base-cased tokenizer. The A100 GPU is found to have a significant

reduction in CO2 emissions and model training time when compared to the T4 GPU. The A100 GPU shortened the model training time by 62.6% on average. Across all five models, the A100 GPU reduced carbon emissions by an astounding 83% on average. It significantly decreased CO2 emissions for BERT, DistilBERT, and the T4 model by 99.7%, 62.7%, and 99.5%, respectively. [3]

The A100 GPU demonstrated how it can significantly enhance CO2 emission efficiency while maintaining the models' overall performance, highlighting the major advantages of using faster hardware for environmental sustainability and reducing training times.

CONCLUSION

As previously stated, the challenge question for which we were seeking a solution was "How to reduce the environmental impacts of large language models without compromising service quality?" This literature research helped us comprehend the seriousness of the situation, which is that while these fast-developing large language models are beneficial to users, they are having a negative impact on the environment due to their carbon footprint. This analysis presented a number of studies in which various ways were evaluated to reduce carbon emissions from these models. We discussed how knowledge distillation, generation directives, parameter-efficient fine tuning, and other approaches have been used to propose more efficient and green models, as well as how these methods influence the process, carbon release, and output of various large language models such as Sprout, BERT, DistilBert, T5, and others. Aside from this, there are many other aspects that contribute to environmental degradation, such as hardware, model architecture, and model training procedures, which can be explored for future research and development, as previously stated. Last but not least, aside from these technical elements, there are additional areas that we can focus on for sustainable large language models such as the role and responsibility of AI developers and corporations, and by paying attention to all of these aspects, we can make progress towards more effective and environmentally friendly large language model development.

DISCUSSION

Although we discussed a variety of strategies for the sustainable growth of AI models, there are many more techniques that could be useful in this research field.

There are other efficient training approaches that can be used but have not been extensively researched, such as data and model parallelism, pipeline parallelism, employing better optimizers, etc. So far, no study has thoroughly and comprehensively assessed the efficiency of these strategies. It is recommended that researchers explore these strategies, evaluate their effectiveness, and then propose new training acceleration methods. To attain an desired efficiency, unique compression algorithms targeted to these extremely large LLMs are urgently required. Furthermore, while knowledge distillation has been the primary focus of current compression efforts for green AI model solutions, other strategies such as quantization and pruning have been given relatively less attention, with fewer research published. While several previous studies have explored dynamic inference approaches to accelerate LLM, there is still lack of thorough research in this field. For example, cascade inference algorithms, which organize tiny and large LLMs in a cascading manner and adaptively select suitable ones rather than depending simply on slower and massive LLMs for each query, could potentially be studied.

In addition to technical procedures, industrial methods such as the role of Tech companies and AI programmers must be addressed. They play a critical role in advancing environmentally friendly AI development. It is essential that they take the initiative in lowering the environmental impact of LLMs. [2]

Tech businesses must acknowledge their contribution to climate change and actively explore solutions to reduce their environmental impact. Implementing environmentally responsible procedures should be a corporate focus. AI developers must be transparent

about the energy and resource usage of their models and activities. Publicly reporting energy usage information promotes accountability and encourages businesses to work for greater sustainability. Competitions and collaborative initiatives focusing on eco-friendly AI development may encourage innovation and inspire the AI community to work together to address this global challenge. [8]

Additionally, Governments can promote environmentally friendly AI by offering research grants, tax breaks, and funding for programs that prioritize energy efficiency. Introducing certifications and standards to measure and regulate AI models' energy consumption will encourage organizations to adopt environmentally responsible practices. Also, Raising the public awareness of AI's environmental impacts may result in increased demand for eco-friendly AI technologies and practices. As a conclusion, by implementing sustainable practices early in AI research, we can ensure a balance between AI advancement and safeguarding the environment.

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